

Mohammed Saferi Rahman

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I am a Computer Science PhD student working at the intersection of HCI, Ubiquitous Computing, and applied LLMs. My research builds theory-grounded, multimodal pipelines that connect individual-level signals to organizational and systemic patterns. I am drawn to research that translates fine-grained interaction data into actionable insights for organizational and institutional design.

Education

Florida State University

PhD in Computer Science (In Progress)

Tallahassee, FL, United States

2022 – 2027 (expected)

MS in Computer Science (Dec, 2024)

- Advisor: Dr. Te-Yen Wu • Lab Name: MakeX • Research Area: HCI, UbiComp, AI
- Current CGPA: 3.77

BRAC University

BS in Computer Science

Dhaka, Bangladesh

2016 – 2020

- CGPA: 3.55
- Graduated with Distinction • Dean's List: Fall '19, Spring '19

Technical & Personal Skills

- **Research Methods:** Mixed-methods study design, In-the-wild deployment (IRB), Qualitative coding, Inter-rater reliability (Krippendorff's α , Cohen's κ), Regression analysis
- **Programming Languages:** *Highly Proficient* Python, C/C++, Java, JavaScript, R, LaTeX
- **ML & Data Frameworks:** PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, OpenAI / Anthropic APIs, OpenCV, MediaPipe, DeepFace, Pandas, NumPy, Matplotlib, SciPy
- **Prototyping & Hardware:** FDC2214 capacitive sensing, MSP430 / Arduino firmware, Real-time signal-processing pipelines, React Native + Firebase (mobile sensing apps), Figma (interaction design)

Publications

- ★ "Double Deep Q-Learning and Faster R-CNN-based Autonomous Vehicle Navigation and Obstacle Avoidance in Dynamic Environment," In *Sensors Journal*, 2021. [\(DOI\)](#)
- "Escape Rooms in Nursing Education: How GPA, Collaborative Problem Solving, and Gameful Experience are related," In *Proceedings of the 18th International Conference of the Learning Sciences-ICLS*, 2024. [DOI](#)
- "Reinforcement Learning based Autonomous Vehicle for Exploration and Exploitation of Undiscovered Track," In proceeding of the *International Conference on Information Networking (ICOIN), IEEE Xplore, Spain*, 2020. [DOI](#)
- "Autonomous Surface Vehicle for Real-time Monitoring of Water Bodies in Bangladesh," In proceeding of the *Global OCEANS Conference, IEEE, Singapore - US Gulf*, 2020. [\(DOI\)](#)

Research Experience

- **JDR-LLM: Inferring Job Demands and Resources of PhD Students Using Multimodal LLM-Based Framework**
 - Publication: Manuscript in preparation (target: CHI'27). Preprint available upon request.
 - Developed a multimodal LLM framework to infer PhD students' weekly JDR scores by analyzing mobile sensor data and digital interaction data through structured two-stage prompting and personalized calibrations.
 - Conducted a four-week in-the-wild study with 14 PhD students, integrating custom smartphone sensing and institutional data pipelines to model perceived stress and academic support.
 - Achieved high alignment with self-reported well-being scores (e.g., MAE $\approx$ 0.35, Spearman $\rho \approx$ 0.88), demonstrating the feasibility of continuous, passive monitoring of PhD students' well-being using LLMs.
- **Emotion-Aware Virtual Patient System for high-fidelity clinical communication training** [\(codebase\)](#)
 - Designed and implemented an LLM based, emotion-aware Virtual Patient (VP) system using Unity, featuring neural TTS, viseme-synchronized facial animations, and configurable affect dynamics to simulate realistic clinical scenarios.
 - Developed a multimodal perception stack combining transformer ASR, prosodic analysis (F0, energy, speaking rate), and vision features (MediaPipe/OpenCV, DeepFace) to infer latent interaction states for adaptive dialogue generation.
 - Integrated a persona-conditioned LLM dialogue policy with real-time coaching, safety constraints, and SOFTEN-based nonverbal feedback dashboards, enabling high-fidelity clinical communication training and planned comparative evaluations against actor-based baselines.
- **So-Shirt: Capacitive-Sensing Garment for Dyadic Social Interaction Recognition**
 - Designed and fabricated a body-scale capacitive-sensing garment with 12 dual-layer textile electrodes (4 torso, 8 arm) driven by a

- 3x FDC2214 array on a custom octagonal PCB, using 3-state multiplexing to separate inter-body coupling from wearer self-coupling for privacy-preserving, single-side dyadic sensing.
- Built an end-to-end real-time acquisition stack — Arduino Nano firmware with dynamic electrode switching and a synchronized Python pipeline recording 12-channel capacitance streams alongside overhead video, producing a labeled dataset of close-proximity interactions.
- Developing a dual-stream deep learning pipeline (Self-Motion Filter with FiLM-conditioned interaction encoder) that uses PHALP + Detectron2-derived SMPL pose targets to jointly reconstruct 13 upper-body joints of the interaction partner and classify dyadic interaction categories from garment signals alone. (*Work in progress*)
- o **Text2IMU: Synthesizing Inertial Sensor Signals from Natural-Language Activity Descriptions** ([codebase](#))
 - Developed a framework that generates realistic inertial sensor signals from natural-language activity descriptions, enabling data augmentation and rapid prototyping for human activity recognition (HAR).
 - Designed a prompt-engineered LLM pipeline to parse activity semantics and synthesize time-series IMU data, with configurable parameters for sampling rate, motion intensity, and temporal patterns.
 - Evaluated generated signals against real IMU recordings, demonstrating their utility for training and benchmarking activity recognition models without costly data collection.
- o **Double Deep Q-Learning and Faster R-CNN-based Autonomous Vehicle Navigation and Obstacle Avoidance in Dynamic Environment**
 - Publication: Sensors Journal, 2021. ([DOI](#))
 - Developed an autonomous vehicle navigation framework that fuses Double Deep Q-Learning (DDQN) for decision-making with Faster R-CNN for real-time object detection and classification in dynamic environments.
 - Implemented the system using Keras and TensorFlow, trained object detectors on the Open Image V5 dataset, and optimized DDQN hyperparameters for lane changing, acceleration, and obstacle avoidance.
 - Validated the integrated model in a GTA V game engine simulator, achieving 94% object detection accuracy and demonstrating precise navigation and obstacle avoidance compared to baseline Q-learning methods.

Notable Projects

- o **ParkNet: Context-Aware Travel Itinerary Generation from Real-Time Signals** ([codebase](#))
 - Designed a web-based platform to help travelers discover and plan personalized trips to U.S. national parks through flexible itineraries, real-time information, and conservation-focused content.
 - Conducted comprehensive user research to identify travel challenges and preferences, informing platform features such as budget-conscious recommendations, itinerary customization, and sustainability integration.
 - Developed mid-fidelity prototypes in Figma, illustrating key user flows including personalized suggestions, group itinerary accommodation, and dynamic travel updates.
- o **Generalizing University Ranking System from websearch** ([codebase](#))
 - Developed a comprehensive university ranking system that integrates data from multiple global ranking sources (Unirank, Webometrics, QS, etc.) to create a unified dataset covering thousands of institutions worldwide.
 - Implemented data collection, parsing, and imputation pipelines using Python, web scraping, and collaborative filtering (e.g., NMF) to handle missing rankings and merge heterogeneous datasets into a single structured CSV.
 - Containerized the entire system with Docker for reproducible deployment, enabling users to run the ranking pipeline end-to-end and generate imputed and combined global ranking tables.
- o **Autonomous Surface Vehicle for Real-time Monitoring of Water Bodies in Bangladesh**
 - Designed and built a GPS-guided autonomous surface vehicle equipped with pH, turbidity, and temperature sensors, enabling real-time water quality monitoring across multiple preset locations.
 - Integrated GSM and Wi-Fi communication modules (SIM900A, ESP8266) for live data transmission to cloud servers, along with a pumping system for on-board sample collection and lab analysis.
 - Deployed and tested the system in five major lakes in Dhaka, successfully collecting and analyzing spatial water quality data to assess pollution levels and environmental trends.

Presentations & Talks

- o **Poster Presentation:** "So-Shirt: Sensing Close-proximity Dyadic Interactions Using Body-scale Capacitive-sensing Garments." *Computer Science Research Expo (CSEXPo)*, Florida State University, 2026.
- o **Poster Presentation:** "Virtual Patient System for Training Medical Students." *Computer Science Research Expo (CSEXPo)*, Florida State University, 2026.
- o **Research Talk:** "JDR-LLM: Inferring Doctoral Job Demands and Resources from Multimodal Traces." *Computer Science Research Expo (CSEXPo)*, Florida State University, 2025.
- o **Poster Presentation:** "Reinforcement Learning based Autonomous Vehicle for Exploration and Exploitation of Undiscovered Track." *International Workshop on Computer Vision and Intelligent Systems (IWCVIS)*, BRAC University, 2019. (**Best Poster Paper Award**)

Relevant Courses

- **Graduate:** Human-Computer Interaction (HCI), Advanced Computer Vision, Introduction to Data Science, Advanced Data Mining, Analytic Methods, Data and Computer Communication, Concurrent, Parallel, and Distributed Programming, Theory of Automata and Formal Languages, Computer & Network System Administration.
- **Undergraduate:** Artificial Intelligence, Neural Networks, Algorithms, Data Structures, Operating Systems, Computer Architecture, Database Systems, Digital System Design, Software Engineering, Probability & Statistics, Linear Algebra, Differential Equations.

Awards & Accolades

- **Research Excellence Award | 2021**
BRAC University, Dhaka
Awarded by the VC
- **Best Poster Paper Award | Dec, 2019**
International Workshop on Computer Vision & Intelligent Systems (IWCVIS)
- **Champion - Programming Contest | 2018**
BRAC University, Dhaka
Inter Department Programming Contest
- **Runner Up - Project Showcasing | 2017**
Islamic University of Technology, Dhaka
IUT 9th ICT Fest
- **Merit Based Scholarship | 2016 - 2020**
BRAC University, Dhaka
70% tuition fee waiver
- **Certificate of Excellence (National) | SSC '13 & HSC '15**
Prothom Alo, Dhaka
Country's Highest Achievers Award